A Conditional Criterion for Twin-Prime Abundance

A source-explicit endgame, classical support estimates, and two optional analytic reductions

Revised mathematical note for independent review

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Abstract

We specify a Ford–Maynard-derived divisor weight, a fixed rough comparison, and the complete interior/exterior partition. We prove an exact identity relating the twin-prime detector to these quantities. A general budget theorem then converts a positive net signed margin into a lower bound of order $X/(\log X)^2$ for twin primes in each sufficiently large dyadic interval. For the literal comparison used here, we supply proofs of its prime-input mass and the complete physical Type-I estimate from classical Bombieri–Vinogradov and elementary sieve truncation. The remaining interior geometry estimates and the decisive exterior bound are stated as explicit hypotheses, not as certified results.

Two optional routes are separated from the main theorem. The degree-twelve route is a proved implication from a rough-input transfer, a moment bound, and a signed estimate. The balanced A5 route is a conditional covariance criterion with an explicit reconstruction budget and a separately retained non-A5 remainder. We repair the smoothing argument by using fixed collars and successive limits, rather than assuming an unproved uniform shrinking-collar asymptotic. No exterior estimate, prime-pair covariance bound, or unconditional twin-prime theorem is proved.

The logical claim of this manuscript

For one fixed source and one fixed $\varepsilon > 0$, with net budget $\kappa = c - d - s > 0$,

$$\begin{gathered}\text{interior comparison lower bound}\\ \text{+ positive interior leakage upper bound}\\ \text{+ complete signed exterior upper bound}\\ \Downarrow\\ Q_X \geq (\kappa + o(1))B_*(X), \quad \kappa > 0\\ \Downarrow\\ \liminf_{X\to\infty} \frac{(\log X)^2}{X}\#\{X/2 < p \leq X : p, p+2 \in \mathbb{P}\} \geq C_2\kappa > 0.\end{gathered}$$

The arrows are proved below. The first three analytic bounds are not all established. The optional marker and A5 routes are not mandatory premises.

Provenance. This revision starts from the supplied 20-page *Twin_Primes_Conditional_Reduction_2026-10-27(2).pdf*. It preserves its arithmetic source. It distinguishes deductions proved in this note, standard external theorems, explicit unproved hypotheses, and project claims not imported as theorems. No independent human referee or formal-kernel certification is asserted.

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Reading guide

Read Sections 1–2 for the complete logical endgame. Section 3 proves the two classical supporting estimates. Section 4 states the remaining interior input. Sections 5 and 6 are alternative research routes. Section 8 lists precisely what still has to be proved. The appendices are not prerequisites for understanding the main implication.

Notation discipline. X is a scale parameter, not a putative twin prime. A prime p is counted only when $p+2$ is also prime. $\mathcal{B}_X(n)$ is a pointwise comparison; $B_*(X)$ is its sum over prime inputs. A nonzero von Mangoldt value can also come from a proper prime power; that contribution is removed explicitly.

1 The fixed arithmetic source

All logarithms are natural. Fix $X > e^2$, write $L = \log X$, and let

$$\mathcal{I}_X = \{n \in \mathbb{N} : X/2 < n \le X,\ n \text{ odd}\}, \qquad \mathcal{P}_X = \mathcal{I}_X \cap \mathbb{P}.$$

Every ε below is fixed before $X \to \infty$. Constants may depend on this fixed ε. No uniformity as $\varepsilon \downarrow 0$ and no effective onset are asserted unless expressly stated.

1.1 Comparison and prime detector

Set

$$z = \lfloor e^{\sqrt{L}} \rfloor, \quad P_z = \prod_{3 \le p \le z} p, \quad V_0(z) = \frac{1}{2} \prod_{3 \le p \le z} \left(1 - \frac{1}{p}\right), \quad \mathcal{B}_X(n) = \frac{\mathbf{1}_{n \in \mathcal{I}_X} \mathbf{1}_{(n+2, P_z)=1}}{V_0(z)}. \tag{1.1}$$

On $\mathcal{I}_X$ put $\Delta_X(n) = \Lambda(n+2) - \mathcal{B}_X(n)$, and define

$$Q_X = \sum_{p \in \mathcal{P}_X} \Lambda(p+2), \qquad B_*(X) = \sum_{p \in \mathcal{P}_X} \mathcal{B}_X(p), \qquad N_2(X) = \#\{p \in \mathcal{P}_X : p+2 \in \mathbb{P}\}. \tag{1.2}$$

The comparison tests only the absence of small prime factors of $n+2$. It does not test primality. Define the positive convergent Euler product

$$C_2 = \prod_{p > 2} \left(1 - \frac{1}{(p-1)^2}\right) > 0. \tag{1.3}$$

Section 3 proves $B_*(X) \sim C_2 X/L$ without a twin-prime asymptotic.

1.2 The precise divisor weight

Use the same shrunken weight as the supplied manuscript:

$$\nu_0 = \frac{16623}{100000}, \quad 0 < \varepsilon \le \frac{\nu_0}{100}, \quad \gamma = \frac{1}{2} - \varepsilon, \quad \sigma = \nu_0 - 2\varepsilon, \quad \delta_g = \frac{1}{2} - 2\varepsilon.$$

For a finite vector $\mathbf{t}$, $|\mathbf{t}|$ denotes its coordinate sum. The endpoint function, extended by zero, is

$$g_0(\mathbf{t}) = \begin{cases} 1, & \mathbf{t} = \varnothing, \\ -1, & \dim \mathbf{t} \in \{1,3\},\ t_i > \nu_0\ (\forall i),\ |\mathbf{t}| \le 1/2, \\ 1, & \dim \mathbf{t} = 2,\ t_i > \nu_0\ (\forall i),\ (1-\nu_0)/2 < |\mathbf{t}| < 1/2, \\ 0, & \text{otherwise.} \end{cases} \tag{1.4}$$

This is the function in the proof of Ford–Maynard Theorem 2.7(b), equation (8.5), with the source conventions specified here [1, pp. 65–67]. Put $g_\varepsilon(\varnothing) = 1$ and $g_\varepsilon(\mathbf{t}) = g_0(\mathbf{t}) \mathbf{1}_{|\mathbf{t}| \le \delta_g}$ for nonempty vectors. Split complete prime-power factors by their prime bases:

$$a(n) = \prod_{p < n^\sigma} p^{\nu_p(n)}, \qquad b(n) = \prod_{p \ge n^\sigma} p^{\nu_p(n)}.$$

For $v = q_1 \cdots q_j$ with multiplicity, write $\mathbf{v}(v;n) = (\log q_1/\log n, \dots, \log q_j/\log n)$ and $\mathbf{v}(1;n) = \varnothing$. Define

$$H_\varepsilon(n) = \sum_{v \mid b(n)} g_\varepsilon(\mathbf{v}(v;n)) \sum_{\substack{u \mid a(n) \\ uv \le n^\gamma}} \mu(u). \tag{1.5}$$

Lemma 1.1. *For every $n > 1$, $|H_\varepsilon(n)| \le \tau(n)$, and $H_\varepsilon(p) = 1$ for every prime p.*

Proof. The map $(u,v) \mapsto uv$ is injective because $(a(n), b(n)) = 1$. Each summand has modulus at most one. For prime p, only $u = v = 1$ survives. □

1.3 Interior, exterior, and the four budget quantities

For $P_\varepsilon = (\gamma, \varepsilon, \sigma)$ let $\mathscr{R}(P_\varepsilon)$ be the set of finite vectors $\mathbf{x}$ satisfying

$$0 < x_i < 1 - \gamma, \quad \sum_i x_i = 1, \quad \sum_{i \in J} x_i \notin [\varepsilon, \varepsilon + \sigma] \quad (\varnothing \neq J \subsetneq \{1, \dots, k\}).$$

A *coagulation* partitions the coordinates into nonempty groups and replaces each group by its sum. Denote the set of coagulations by $\mathscr{C}(\mathscr{R}(P_\varepsilon))$. This fixes the definitions used in [1, Definitions 4.3 and 4.5]. Set

$$\mathcal{N}_{\varepsilon,X} = \{n \in \mathcal{I}_X : n \text{ composite}, \ \mathbf{v}(n; n) \in \mathscr{C}(\mathscr{R}(P_\varepsilon))\}, \qquad \mathcal{E}_{\varepsilon,X} = \mathcal{I}_X \setminus (\mathcal{P}_X \cup \mathcal{N}_{\varepsilon,X}).$$

Prime factors in $\mathbf{v}(n; n)$ are counted with multiplicity. Finally,

$$T_X = \sum_{n \in \mathcal{I}_X} H_\varepsilon(n)\Delta_X(n), \qquad S_X = \sum_{n \in \mathcal{E}_{\varepsilon,X}} H_\varepsilon(n)\Delta_X(n), \tag{1.6}$$

$$C_X = B_*(X) + \sum_{n \in \mathcal{N}_{\varepsilon,X}} H_\varepsilon(n)\mathcal{B}_X(n), \qquad D_X = \sum_{n \in \mathcal{N}_{\varepsilon,X}} (H_\varepsilon(n))_+ \Lambda(n+2). \tag{1.7}$$

Here $x_+ = \max(x, 0)$. D_X is not small by definition. It is the positive interior leakage that must be estimated. All four quantities use the same ε, source, comparison, and physical interval.

2 The conditional theorem and its complete proof

Lemma 2.1 (Exact partition and direction of the inequality)**.** *One has*

$$Q_X = C_X + T_X - S_X - \sum_{n \in \mathcal{N}_{\varepsilon,X}} H_\varepsilon(n)\Lambda(n+2) \ \geq \ C_X + T_X - S_X - D_X. \tag{2.1}$$

Proof. Partition the full sum T_X into prime inputs, interior composites, and exterior composites. Lemma 1.1 makes the prime-input part $Q_X - B_*(X)$. Expanding Δ_X on the interior gives

$$T_X = Q_X - B_*(X) + \sum_{\mathcal{N}_{\varepsilon,X}} H_\varepsilon \Lambda(n+2) - \sum_{\mathcal{N}_{\varepsilon,X}} H_\varepsilon \mathcal{B}_X + S_X.$$

Rearrange and use $H_\varepsilon \Lambda \leq (H_\varepsilon)_+ \Lambda$. □

In particular, an *upper* bound for S_X is helpful; no absolute-value bound is required. Negative exterior contributions improve the budget.

Lemma 2.2 (Removal of proper prime powers)**.** *For sufficiently large X,*

$$Q_X \leq \log(X+2) N_2(X) + O(X^{1/2} L^2). \tag{2.2}$$

Proof. The number of values $p^a \leq X + 2$ with $a \geq 2$ is at most $\sum_{2 \leq a \leq \log_2(X+2)} (X+2)^{1/a} = O(\sqrt{X}L)$. Their total von Mangoldt weight is $O(\sqrt{X}L^2)$. All other nonzero terms in Q_X are genuine twins, with weight at most $\log(X+2)$. □

Theorem 2.3 (General conditional budget theorem)**.** *For the source above, suppose that fixed constants $b > 0$, $c \in \mathbb{R}$, $t, d \geq 0$, and $s \in \mathbb{R}$ satisfy, for all sufficiently large X,*

$$B_*(X) \geq bX/L, \quad C_X \geq cB_*(X), \quad T_X \geq -tB_*(X), \quad D_X \leq dB_*(X), \quad S_X \leq sB_*(X). \tag{2.3}$$

If $\kappa = c - t - d - s > 0$, then

$$Q_X \geq \kappa B_*(X), \qquad N_2(X) \geq \frac{\kappa b}{4} \frac{X}{L^2}$$

for every sufficiently large X. In particular there are infinitely many twin primes.

Proof. The first conclusion is immediate from (2.1). In (2.2), choose X sufficiently large that the proper-power error is at most $\kappa bX/(2L)$ and that $\log(X+2) \le 2L$. This gives the displayed lower bound. Given any M, choose such an $X > 2M$; its positive lower bound supplies a twin pair whose smaller member exceeds M. □

Theorem 2.4 (Asymptotic form, with no artificial loss of the leading constant)**.** *Suppose $B_*(X) \sim b_*X/L$ with $b_* > 0$ and*

$$\liminf \frac{C_X}{B_*(X)} \ge c, \qquad \liminf \frac{T_X}{B_*(X)} \ge -t,$$
$$\limsup \frac{D_X}{B_*(X)} \le d, \qquad \limsup \frac{S_X}{B_*(X)} \le s, \qquad \kappa = c - t - d - s > 0.$$

Then

$$\liminf_{X\to\infty} \frac{L^2 N_2(X)}{X} \ge b_*\kappa. \tag{2.4}$$

Proof. The elementary inequalities for liminf and limsup, applied to (2.1), give $\liminf Q_X/B_*(X) \ge \kappa$. Multiply (2.2) by L/X. The error becomes $O(X^{-1/2}L^3) = o(1)$, and $\log(X+2)/L \to 1$. This proves (2.4). □

2.1 What the original numerical budgets actually require

The uploaded note assumes

$$C_X \ge \frac{7c_F}{8}B_*(X), \quad |T_X| \le \frac{c_F}{16}B_*(X), \quad D_X \le \frac{c_F}{16}B_*(X), \qquad c_F > 0.$$

Consequently the sufficient exterior threshold is not intrinsically $c_F/4$. The following are three distinct statements.

Available supporting information	Sufficient exterior condition
Only the displayed finite budgets	$\limsup S_X/B_*(X) < 3c_F/4$
Same C_X, D_X budgets and $T_X = o(B_*(X))$	$\limsup S_X/B_*(X) < 13c_F/16$
$C_X/B_*(X) \ge c_F - K_G\varepsilon - o(1)$; $D_X/B_*(X) \le K_D\varepsilon + o(1)$; $T_X = o(B_*(X))$	$\limsup S_X/B_*(X) < c_F - (K_G + K_D)\varepsilon$

Each inequality must have a fixed positive gap and must use the same fixed ε. These are sufficient thresholds, not necessary conditions for twin primes and not optimality statements about every analytic method. The old $c_F/4 - \rho$ target remains a stronger valid target.

Corollary 2.5 (Joint-source certificate)**.** *If $B_*(X) \sim C_2X/L$, $T_X = o(B_*(X))$, and*

$$\liminf_{X\to\infty} \frac{C_X - D_X - S_X}{B_*(X)} > 0,$$

then $N_2(X) \gg X/L^2$. No separate estimate $S_5 = o(B_*(X))$ is needed.

Proof. Use (2.1) and Lemma 2.2. □

This corollary permits joint cancellation, but is only a logical criterion. Writing it down does not estimate its left side.

3 Two supporting estimates from classical distribution

This section fills in two proof obligations that were only described in the uploaded note. The external input is classical maximal Bombieri–Vinogradov; neither an unproved prime-pair conjecture nor a new Type-II estimate is used.

3.1 External input and divisor weights

For fixed $\delta > 0$ and every $A > 0$, maximal BV gives

$$\sum_{q \le X^{1/2-\delta}} \max_{(a,q)=1} \sup_{J \subseteq [1,2X]} \left| \sum_{\substack{n \in J \cap \mathbb{N} \\ n \equiv a \ (q)}} \Lambda(n) - \frac{|J|}{\varphi(q)} \right| \ll_{A,\delta} XL^{-A}. \tag{3.1}$$

This is the fixed-power-margin version of the standard theorem; arbitrary intervals follow by subtracting two maximal prefixes [2, Theorem 17 and its maximal variants]. For every fixed $C \ge 0$, a factor $\tau(q)^C$ can be inserted at the cost of requesting stronger logarithmic precision. Indeed, if $E(q)$ denotes the maximal error, then

$$\sum_q \tau(q)^C E(q) \le \left(\sum_q E(q) \right)^{1/2} \left(\sum_q \tau(q)^{2C} E(q) \right)^{1/2}.$$

The trivial bound $E(q) \ll XL\tau(q)/q$ and the fixed divisor moment $\sum_{q \le X} \tau(q)^u/q \ll_u L^{O_u(1)}$ bound the second factor. Choosing the exponent in (3.1) sufficiently large proves the claim. All weights and exponents here are fixed before X.

Proposition 3.1 (Prime-input comparison mass)**.** *For the literal comparison* (1.1)*,*

$$B_*(X) = (C_2 + o(1))X/L. \tag{3.2}$$

Proof. Put $W(z) = \prod_{3 \le p \le z}(1 - 1/(p-1))$. Adjacent even and odd Bonferroni sums approximate $\mathbf{1}_{(p+2,P_z)=1}$. At truncation order $K = O_A(\log L)$ all retained squarefree moduli satisfy

$$s \le z^{K+1} = \exp(O_A(\sqrt{L}\log L)) = X^{o(1)}.$$

For each such s, prime inputs satisfy $p \equiv -2 \pmod{s}$, a unit class. BV in prime-counting form, obtained from (3.1) by partial summation, controls the *sum* of progression errors over these moduli by $O_A(XL^{-A})$. No modulus-count multiplier is added. The reciprocal main-term truncation error is at most

$$\frac{\left(\sum_{3 \le p \le z} 1/(p-1)\right)^{K+1}}{(K+1)!}.$$

Since the numerator's prime sum is $\frac{1}{2}\log L + O(1)$, a sufficiently large fixed constant in K makes this smaller than every requested fixed negative power of L. Thus

$$\#\{p \in \mathcal{P}_X : (p+2, P_z) = 1\} = W(z)\big(\mathrm{Li}(X) - \mathrm{Li}(X/2)\big) + O_A(XL^{-A}).$$

At every odd prime,

$$\frac{1 - 1/(p-1)}{1 - 1/p} = 1 - \frac{1}{(p-1)^2}.$$

Consequently $W(z)/V_0(z) = 2C_2(z) \to 2C_2$. Mertens' theorem gives $V_0(z)^{-1} \ll \sqrt{L}$, and $\mathrm{Li}(X) - \mathrm{Li}(X/2) \sim X/(2L)$. Division by $V_0(z)$ proves (3.2). □

3.2 A weighted progression lemma with the literal comparison

For an odd integer d, define

$$\rho_z(d) = \frac{1}{d} \prod_{p\mid d,\ 3\le p\le z} (1-1/p)^{-1} = \frac{1}{\varphi(d)} \prod_{p\mid d,\ p>z} (1-1/p). \tag{3.3}$$

This formula holds for nonsquarefree d as well.

Lemma 3.2 (Subcritical weighted Type I)**.** *Fix $\delta > 0$ and $B, C \ge 0$. For each odd $d \le X^{1/2-\delta}$ let f_d be a complex bounded-variation function on $[X/2, X]$ satisfying*

$$\|f_d\|_\infty + \mathrm{Var}(f_d) \le L^B \tau(d)^C.$$

Then, for every $A > 0$,

$$\sum_{\substack{d\le X^{1/2-\delta}\\ d\ \mathrm{odd}}} \sum_{\substack{n\in I_X\\ d\mid n}} f_d(n)\Delta_X(n) \ll_{A,B,C,\delta} XL^{-A}. \tag{3.4}$$

The functions f_d need not factor, but their actual variation is part of the hypothesis. An arbitrary exterior indicator is not licensed.

Proof. For the prime term, $d \mid n$ with n odd is equivalent to $n + 2 \equiv d + 2 \pmod{2d}$, a reduced residue class. Its main density per unit interval length is $1/\varphi(d)$. Weighted maximal BV and partial summation handle all f_d simultaneously.

For the comparison term, a prime $p \mid d$ cannot divide $n + 2$. The remaining sieve divisors $s \mid P_z$ satisfy $(s, d) = 1$. CRT gives

$$\#\{n \in J : n \text{ odd},\ d \mid n,\ s \mid n+2\} = \frac{|J|}{2ds} + O(1)$$

uniformly over real intervals $J \subseteq (X/2, X]$. Truncate the Bonferroni expansion at $K = O_{A,B,C}(\log L)$. After division by $V_0(z)$, the main term is $|J|\rho_z(d)$. The factorial reciprocal tails, summed with $\tau(d)^C L^B/d$, are $O_A(XL^{-A})$. The entire lattice remainder, including all d, is at most

$$X^{1/2-\delta} \exp(O_{A,B,C}(\sqrt{L}\log L)) L^{O_{B,C}(1)} = O_A(XL^{-A}).$$

This uses the actual number of truncated moduli; it is not hidden inside an unbounded $X^{o(1)}$ at the terminal scale.

Finally,

$$0 \le \frac{1}{\varphi(d)} - \rho_z(d) \le \frac{1}{\varphi(d)} \sum_{p\mid d,\ p>z} \frac{1}{p} \ll \frac{L}{z\log z\,\varphi(d)}.$$

The sum $\sum_{d\le X} \tau(d)^C/\varphi(d)$ is a fixed power of L. Thus the total main-density mismatch is $XL^{O_{B,C}(1)}/z$, smaller than every fixed XL^{-A}. Partial summation transfers the uniform interval estimates to f_d and completes the proof. □

Proposition 3.3 (Complete physical Type I for the specified weight)**.** *For fixed $\varepsilon > 0$ and every $A > 0$,*

$$T_X \ll_{A,\varepsilon} XL^{-A}. \tag{3.5}$$

Proof. For fixed d and real $t \in [X/2, X]$, split the full prime powers of d at $p < t^\sigma$ into $d_1(t)d_2(t) = d$, and put

$$f_d(t) = \mathbf{1}_{d\le t^\gamma} \mu(d_1(t)) g_\varepsilon(\mathbf{v}(d_2(t); t)).$$

Then $H_\varepsilon(n) = \sum_{d\mid n} f_d(n)$ and $d \le X^\gamma$ on the support. The split changes only at the $O(\omega(d))$ thresholds $t = p^{1/\sigma}$. Between consecutive thresholds, d_1, d_2 are fixed. The nonzero part of (1.4), including the

truncation and $d \le t^{\gamma}$, is then an intersection of a fixed number of intervals in t. Its value is 0, 1, or -1. If $\Omega(d_2) > 3$ it is identically zero. Thus

$$\|f_d\|_\infty + \mathrm{Var}(f_d) \ll 1 + \omega(d) \ll L.$$

Apply Lemma 3.2 with $\delta = \varepsilon$. This estimates the full physical sum only; it does not insert $\mathbf{1}_{\mathcal{E}_{\varepsilon,X}}$ into the lemma. □

Remark 3.4 (What has actually been advanced). Propositions 3.1 and 3.3 now have explicit classical proofs in this manuscript, rather than references only to private project reports. They are not new prime-correlation estimates. They justify $b_* = C_2$ and $t = 0$ in Theorem 2.4; they do not bound S_X.

4 The remaining interior input

The published endpoint geometry and the current shrunken arithmetic source must not be identified without a proof. The following hypothesis states exactly the remaining supporting information used by the direct route.

Hypothesis 4.1 (Interior comparison and positive leakage)**.** For the fixed ε and source in Section 1, there are fixed $c > 0$ and $d \ge 0$ such that

$$\liminf_{X\to\infty} \frac{C_X}{B_*(X)} \ge c, \qquad \limsup_{X\to\infty} \frac{D_X}{B_*(X)} \le d. \tag{4.1}$$

The useful case has $c > d$.

Corollary 4.2 (The direct criterion in its shortest useful form)**.** *Assume Hypothesis 4.1. If*

$$\limsup_{X\to\infty} \frac{S_X}{B_*(X)} \le s < c - d, \tag{4.2}$$

then

$$\liminf_{X\to\infty} \frac{L^2 N_2(X)}{X} \ge C_2(c - d - s) > 0.$$

Proof. Apply Theorem 2.4 with Propositions 3.1 and 3.3, so $b_* = C_2$ and $t = 0$. □

Open analytic responsibility. Neither Hypothesis 4.1 nor (4.2) is proved in full here. Their conjunction is a genuine assumption, not a software label.

4.1 Relation to the geometric constant

The limiting constant in the source records is

$$c_F = 1 + \sum_{k=2}^{6} \frac{1}{k!} \int_{\substack{x_i \ge v_0 \\ \sum_i x_i = 1}} \frac{\sum_{J\subseteq\{1,\dots,k\}} g_0(\mathbf{x}_J)}{x_1 \cdots x_k}\, dx_1 \cdots dx_{k-1}. \tag{4.3}$$

The measure is coordinate projection measure, with $x_k = 1 - \sum_{i<k} x_i$. The factor $1/k!$ belongs to this unordered simplex; it does not belong to an ordered A5 box integral. Ford–Maynard report a positive lower bound $6 \cdot 10^{-6}$ in the proof of Theorem 2.7(b) [1, p. 67]. The project's more precise enclosure is not recomputed here and is not needed in the conditional theorem.

A sufficient source-specific interior theorem would be

$$\frac{C_X}{B_*(X)} \ge c_F - K_G\varepsilon - o_\varepsilon(1), \qquad \frac{D_X}{B_*(X)} \le K_D\varepsilon + o_\varepsilon(1), \tag{4.4}$$

with K_G, K_D independent of sufficiently small positive ε. Choose ε once, with $(K_G + K_D)\varepsilon$ below the intended budget margin; then take $X \to \infty$. One cannot tune ε after seeing X without additional uniform estimates.

For example, a decomposition $\mathcal{N}_{\varepsilon,X} = \mathcal{N}_1 \sqcup \mathcal{N}_2$ with $H_\varepsilon \le 0$ on $\mathcal{N}_1$, together with

$$B_*(X) + \sum_{\mathcal{N}_1} H_\varepsilon \mathcal{B}_X \ge (c_F - K_1\varepsilon - o(1))B_*(X),$$
$$\sum_{\mathcal{N}_2} |H_\varepsilon| \mathcal{B}_X \le (K_2\varepsilon + o(1))B_*(X),$$
$$\sum_{\mathcal{N}_2} |H_\varepsilon| \Lambda(n+2) \le (K_3\varepsilon + o(1))B_*(X),$$

would imply (4.4) with $K_G = K_1 + K_2$, $K_D = K_3$. This elementary conversion is proved by adding the entire collar back. The actual partition, its source geometry, repeated factors, comparison measure, and uniform upper-sieve bound remain proof obligations. A small geometric volume alone is not a prime-output theorem.

5 Optional route I: the degree-twelve signed sum

Nothing in Corollary 4.2 requires this route. Its purpose is to replace one complete exterior target by another under two explicit transfer hypotheses. No hidden assertion that the replacement is easier is made.

5.1 The two analytic assumptions

For the same fixed ε, let

$$K_X = \lceil 448 \log L \rceil, \quad Y = X^{\varepsilon/(8K_X)}, \quad \theta = \frac{\log Y}{L}, \quad \mathcal{Q}_Y = \{n \in \mathcal{I}_X : \mu(n)^2 = 1,\ P^-(n) > Y\}.$$

Use $P^-(1) = \infty$.

Hypothesis 5.1 (Complete exterior rough-input transfer)**.** With $S_Y = \sum_{n \in \mathcal{E}_{\varepsilon,X} \cap \mathcal{Q}_Y} H_\varepsilon(n)\Delta_X(n)$,

$$S_X = S_Y + O_\varepsilon(XL^{-4}).$$

This is not a consequence of support inclusion. The claimed source proof requires the growing-cutoff expansion, LCM regrouping, squarefree error, and exterior restoration; no arbitrary masked Type-I theorem is assumed.

Put

$$y = e^{L^{1/10}}, \quad w = e^{L^{1/5}}, \quad \mathcal{P}_X = \{p \in \mathbb{P} : y < p \le w\}, \quad R(n) = \sum_{p \in \mathcal{P}_X} \mathbf{1}_{p \mid n+1}, \quad V = \sum_{p \in \mathcal{P}_X} \frac{1}{p-2}.$$

Mertens' estimate implies $V = \frac{1}{10}\log L + o(1)$. The local denominator is exact: when $p \nmid c$, exclusion of $q \equiv 0, -2c^{-1} \pmod p$ leaves $p-2$ residues, exactly one of which has $cq \equiv -1 \pmod p$. This is a finite CRT model, not independence of two actual primes.

Hypothesis 5.2 (Uniform twelfth moment)**.** For every fixed $c_0 > 0$ and $A > 0$, uniformly for $c_0/\log L \le \theta \le 1/4$ and $Y = X^\theta$,

$$\sum_{n \in \mathcal{Q}_Y} \tau(n)[\Lambda(n+2) + \mathcal{B}_X(n)]|R(n) - V|^{12} \ll_{c_0,A} \frac{XV^6}{L\theta^4} + XL^{-A}. \tag{5.1}$$

The supplied report gives a positive-sieve argument, not an independent certification. This manuscript verifies the deduction *from* (5.1); it does not promote all details of that argument to a proved theorem. In particular, CRT moments alone do not prove the actual weighted estimate over integers. The shrinking θ, full prefix population, Selberg normalization, prime powers, and all lattice errors must be controlled in such a proof.

5.2 The conditional transfer, proved explicitly

Define

$$M_{12}(n) = 1 - \left(1 - \frac{R(n)}{V}\right)^{12}, \qquad S_{12}(X) = \sum_{n \in \mathcal{E}_{\varepsilon,X} \cap Q_Y} H_\varepsilon(n) \Delta_X(n) M_{12}(n).$$

M_{12} can be negative when $R(n) > 2V$. It is not a probability weight.

Proposition 5.3 (Restoration with a paid error)**.** *Assume Hypotheses 5.1 and 5.2. Then*

$$S_X = S_{12}(X) + O_\varepsilon\left(\frac{X}{L(\log L)^2}\right) = S_{12}(X) + o(B_*(X)). \tag{5.2}$$

Proof. By Lemma 1.1,

$$|S_Y - S_{12}| \le V^{-12} \sum_{n \in Q_Y} \tau(n)[\Lambda(n+2) + \mathcal{B}_X(n)]|R(n) - V|^{12} \ll \frac{X}{L\theta^4 V^6} + XL^{-4}V^{-12}.$$

Here $\theta^{-4} \ll_\varepsilon (\log L)^4$ and $V^{-6} \ll (\log L)^{-6}$. Add Hypothesis 5.1 and use Proposition 3.1. □

For $1 \le j \le 12$ set

$$a_j(V) = j! \sum_{k=j}^{12} (-1)^{k+1} \binom{12}{k} \left\{ {k \atop j} \right\} V^{-k},$$

where braces denote Stirling numbers of the second kind, and put

$$\mathcal{D}_X = \{d \ge 2 : \mu(d)^2 = 1,\ p \mid d \Rightarrow p \in \mathcal{P}_X,\ 1 \le \omega(d) \le 12\}.$$

Lemma 5.4 (Exact divisor expansion)**.** *One has*

$$M_{12}(n) = \sum_{\substack{d \in \mathcal{D}_X \\ d \mid n+1}} a_{\omega(d)}(V), \qquad \sum_{d \in \mathcal{D}_X} \frac{|a_{\omega(d)}(V)|}{d} + \sum_{d \in \mathcal{D}_X} \frac{|a_{\omega(d)}(V)|}{\varphi(d)} \ll 1.$$

Every $d \in \mathcal{D}_X$ is at most $\exp(12L^{1/5})$.

Proof. Expand $1 - (1 - R/V)^{12}$ and use $R^k = \sum_{j=1}^k \left\{ {k \atop j} \right\} j! \binom{R}{j}$. The binomial counts the squarefree divisors of $n+1$ with j marked prime factors. For fixed degree, $|a_j(V)| \ll_j V^{-j}$ when $V \ge 1$. The elementary symmetric-sum bound

$$\sum_{\substack{\omega(d)=j \\ d \in \mathcal{D}_X}} \frac{1}{d} \le \frac{1}{j!} \left(\sum_{p \in \mathcal{P}_X} \frac{1}{p} \right)^j$$

and its $1/\varphi(d)$ analogue prove the harmonic bounds. The prime sums are $V + o(1)$. The size bound follows from at most twelve factors $\le w$. □

Thus $S_{12} = \mathcal{J}_{12}$ exactly, where

$$\mathcal{J}_{12}(X) = \sum_{d \in \mathscr{D}_X} a_{\omega(d)}(V) \sum_{\substack{n \in \mathcal{E}_{\varepsilon,X} \cap Q_Y \\ n \equiv -1 \pmod d}} H_\varepsilon(n)[\Lambda(n+2) - \mathcal{B}_X(n)]. \tag{5.3}$$

The small auxiliary modulus and bounded harmonic mass are not estimates of the inner signed sum.

Corollary 5.5 (Correct conditional marker criterion)**.** *Assume Hypotheses 4.1, 5.1, and 5.2. If*

$$\limsup_{X\to\infty} \frac{\mathcal{J}_{12}(X)}{B_*(X)} < c - d,$$

then there are $\gg X/L^2$ twin primes in $(X/2, X]$ for all sufficiently large X.

Proof. Equation (5.2) preserves the strict limsup gap, and Corollary 4.2 applies. □

The historical target $\limsup \mathcal{J}_{12}/B_*(X) \le c_F/8$ remains a valid stronger choice under its historical interior budgets. The assertion itself is still open. Reconstruction error is not cancellation of $\mathcal{J}_{12}$.

6 Optional route II: the current balanced A5 frontier

This section preserves the newer research frontier without importing a claimed Type-II estimate. It is not a prerequisite for the direct theorem, and it does not replace the complete exterior by A5.

6.1 Exact source and the non-A5 remainder

Appendix A gives all 133 original half-open boxes. For $\mathbf{k} = (k_1, k_2, k_3, k_4)$, the first four coordinates satisfy $x_i \in [k_i/100, (k_i+1)/100)$ and $x_5 = 1 - \sum_{i=1}^4 x_i$. Let $\mathcal{A}_5(X)$ be the corresponding products of five distinct ordered primes. Its supplied finite certificate gives $H_\varepsilon = 2$ on 132 boxes and $H_\varepsilon = 4$ on $(12, 15, 17, 25)$. The finite-source verification and exterior membership are explained in Appendix A. Put

$$S_5(X) = \sum_{n \in \mathcal{A}_5(X)} H_\varepsilon(n)\Delta_X(n), \qquad R_5(X) = S_X - S_5(X).$$

All 133 labelled boxes admit

$$q = p_3p_4, \quad m = p_1p_2p_5, \quad \frac{21}{50} \le \frac{\log q}{\log n} < \frac{12}{25}, \quad QM \asymp X. \tag{6.1}$$

The former seven-box correction was a problem of a particular unmasked projection, not an obstruction to (6.1) with all labels retained. This is finite source geometry, not Type-II cancellation.

6.2 A safe smoothing repair: fixed collars, then successive limits

The latest project report asserted a quantitative $O(\delta_L B_*(X))$ collar bound for a shrinking $\delta_L = (\log L)^{-C}$. A fixed-domain asymptotic alone does not prove that uniform estimate. Even if granted, this error is generally larger than $XL^{-1-\eta/2}$, so the latter cannot be reported as the total bound for the sharp source.

The following repair retains the desired $o(B_*(X))$ conclusion without requiring either assertion. Fix a sufficiently small $\delta > 0$ first. Smooth the eight original lower/upper tests from the inside, and call the resulting source sum $S_{5,\delta}$.

Lemma 6.1 (Fixed-collar absolute bound)**.** *There is a fixed C such that*

$$\limsup_{X\to\infty} \frac{|S_5(X) - S_{5,\delta}(X)|}{B_*(X)} \le C\delta. \tag{6.2}$$

Proof. All five prime coordinates are bounded away from zero, and their ordered ranges have fixed separation. For $n \in (X/2, X]$, replacing $\log n$ by L changes each coordinate by $O(1/L)$. Thus for fixed δ, the boundary population has its first-four prefix in a fixed enlargement of finitely many coordinate strips of total volume $O(\delta)$.

Write $n = Pp_5$, with $P = p_1p_2p_3p_4$ fixed, and $N = X/P$. On the enlarged source $N \geq X^a$ for a fixed $a > 0$. A two-linear-form Selberg upper sieve for p_5 and $Pp_5 + 2$ gives $O_a(N/L^2)$ genuine-prime pairs in the full dyadic last-prime interval. Its local singular factors are uniformly bounded here: P has at most four distinct prime factors, each at least X^a after reducing a. This is the standard nonnegative upper-sieve construction [3, 4]; one may take its divisor support $\leq N^{1/10}$, so its $O(N^{1/5}\log^{O(1)} N)$ lattice remainder is absorbed. Multiplying by the output logarithm costs $O(L)$.

For the comparison, apply small-modulus BV to the last prime and use Bonferroni at z. The moduli are $\exp(O(\sqrt{L}\log L)) \ll N^{1/2-\eta}$ for a fixed $\eta > 0$. Division by the literal $V_0(z)$ gives an upper bound $O_a(N/L)$, with errors of arbitrarily high fixed logarithmic order. Prefix restoration uses $\sum 1/P = O(1)$.

More precisely, reciprocal-prime sums in fixed exponent intervals converge to the measure dt/t. The reciprocal measure of the finitely many first-four coordinate strips is therefore $O(\delta) + o_\delta(1)$; the density is bounded on the fixed compact region. Sum the two upper bounds $O(X/(PL))$ over those prefixes. Proper prime-power outputs cost $O(\sqrt{X}L^2)$ because the source weight is bounded. Divide by Proposition 3.1 and take limsup. □

Lemma 6.2 (Separation and true source norms)**.** *For each fixed $\delta > 0$, $S_{5,\delta}$ is, up to arbitrary fixed logarithmic error, a sum/integral of balanced blocks*

$$T(\alpha,\beta) = \sum_{m \bmod d} \beta_m \sum_q \alpha_q \mathbf{1}_{m\in I(q)} \Delta_X(qm), \quad I(q) = J \cap (X/(2q), X/q],$$

with total coefficient variation $O_\delta(L)$ and

$$\|\alpha\|_2^2 \ll Q/L, \qquad \|\beta\|_2^2 \ll M/L. \tag{6.3}$$

The actual separated coefficients are bounded prime convolutions on $\alpha : q = p_3p_4$ and $\beta : m = p_1p_2p_5$, with their disjoint prime ranges and unit-modulus Mellin twists retained.

Proof. Smooth functions should be applied to the linear logarithmic tests $(\log p_i - a_i \log n)/L$ and $(b_i \log n - \log p_i)/L$, not directly Fourier-expanded in a ratio with variable denominator. On the compact source, these tests have transition collars of width $O(\delta)$ in the normalized coordinates. Multiply the smooth test functions by a compact cutoff equal to one on their actual argument range. For fixed δ their Fourier transforms have finite L^1 norm and rapidly decaying tails. Each Fourier phase is an exactly separated product of powers of the individual primes, because $\log n = \sum_i \log p_i$. Truncation at a fixed power of L, chosen for the desired error, has arbitrary logarithmic accuracy. The tail is summed using the elementary fixed divisor moments, not by assuming cancellation.

The product of eight L^1 costs is $O_\delta(1)$. There are $O(L)$ compatible dyadic (Q, M) pairs. A factorization in disjoint prime slots is unique. To count the q support, fix p_3 and use $\pi(2Q/p_3) \ll Q/(p_3L)$; the reciprocal-prime mass in its fixed exponent range is $O(1)$. To count m, fix p_1, p_2 and count p_5 in the same way. This proves (6.3). The constants are uniform under unit-modulus twists. □

6.3 The exact covariance and its evidence boundary

For q_1, q_2 put $J_{12} = I(q_1) \cap I(q_2)$ and $\ell_{12} = |J_{12}|$. Let $\vartheta(u) = \log u\, \mathbf{1}_{u\in\mathbb{P}}$ and

$$P_X(q_1,q_2) = \sum_{\substack{m\in J_{12}\\ m \bmod d}} \vartheta(q_1m+2)\vartheta(q_2m+2).$$

For coprime odd $q_1, q_2 > 1$, define

$$\mathfrak{S}(q_1, q_2) = 2C_2 \prod_{p>2,\ p \mid q_1 q_2 (q_2 - q_1)} \frac{p-1}{p-2}.$$

The product is a specified local model, not a prime-pair theorem. Set

$$\mathscr{E}_X(\alpha) = \sum_{\substack{m \in J \\ m \bmod d}} \left| \sum_q \alpha_q \mathbf{1}_{m \in I(q)} \Delta_X(qm) \right|^2, \tag{6.4}$$

$$\mathscr{V}_X(\alpha) = \sum_{(q_1,q_2)=1} \alpha_{q_1} \overline{\alpha_{q_2}} \big(P_X(q_1, q_2) - \ell_{12} \mathfrak{S}(q_1, q_2) \big). \tag{6.5}$$

Both are real after the full ordered-pair summation; only $\mathscr{E}_X$ is manifestly nonnegative.

Hypothesis 6.3 (Dispersion background adapter)**.** For every fixed $A > 0$, uniformly over the actual intervals and bounded source coefficients in Lemma 6.2,

$$\mathscr{E}_X(\alpha) = \mathscr{V}_X(\alpha) + O_A(XQL^{-A}). \tag{6.6}$$

The supplied balanced-dispersion report contains a written derivation using BV for the two mixed terms and CRT for comparison/comparison. It uses moduli $2q_1 s < X^{1/2-\eta}$, with s composed of comparison primes and q_1 polynomially rough. This manuscript retains that report as an adapter to audit, not an independently certified Type-II result. Shared-prime pairs and prime powers are also part of its stated error.

Lemma 6.4 (An elementary near-diagonal bound)**.** *For every fixed $A > 0$, choosing a sufficiently large fixed C gives*

$$\left| \sum_{\substack{(q_1,q_2)=1 \\ 0 < |q_1 - q_2| \le QL^{-C}}} \alpha_{q_1} \overline{\alpha_{q_2}} \big(P_X(q_1, q_2) - \ell_{12} \mathfrak{S}(q_1, q_2) \big) \right| \ll_A XQL^{-A}.$$

Proof. There are $O(Q/L)$ source values of q_1 and at most $O(QL^{-C})$ integers q_2 in the required range. Trivially $P_X \ll ML^2$. Mertens' product bound gives $\mathfrak{S} \ll L$, by bounding its factors by all odd primes up to $O(Q^3)$. Thus each summand has modulus $O(ML^2)$, and the total is $O(MQ^2L^{1-C}) = O(XQL^{1-C})$. Take $C > A + 2$. No prime-pair asymptotic or uniform two-prime upper sieve is needed here. □

Hypothesis 6.5 (Far prime-pair covariance)**.** For some fixed $\eta > 0$, for every fixed smoothing width $\delta > 0$ and all actual separated coefficients and Mellin twists in Lemma 6.2,

$$\mathscr{V}_X^{\mathrm{far}}(\alpha) \ll_{\delta,\eta} XQL^{-3-\eta}. \tag{6.7}$$

Here "far" is the symmetric restriction $(q_1, q_2) = 1$, $|q_1 - q_2| > QL^{-C}$, with C chosen in Lemma 6.4. This is an open analytic hypothesis.

Proposition 6.6 (Repaired current A5 conditional frontier)**.** *Hypotheses 6.3 and 6.5 imply $S_5(X) = o(B_*(X))$.*

Proof. Fix $\delta > 0$. Choose the background and near-diagonal precision larger than $3 + \eta$. By Cauchy, (6.3), and (6.6), each block satisfies

$$|T(\alpha, \beta)|^2 \le \|\beta\|_2^2 \mathscr{E}_X(\alpha) \ll \frac{M}{L} XQL^{-3-\eta} \ll X^2 L^{-4-\eta}.$$

Thus $T \ll XL^{-2-\eta/2}$. Lemma 6.2 gives $S_{5,\delta} \ll_\delta XL^{-1-\eta/2}$. For this fixed δ, divide by $B_*(X)$ and take $X \to \infty$. Lemma 6.1 gives $\limsup |S_5|/B_*(X) \le C\delta$. Now let $\delta \downarrow 0$. This proves $o(B_*(X))$, but does not assert the same log-power rate for the original sharp sum. □

Corollary 6.7 (A5 is not the complete theorem)**.** *Assume Hypothesis 4.1, both hypotheses of Proposition 6.6, and the additional complete-remainder bound*

$$\limsup_{X\to\infty} \frac{R_5(X)}{B_*(X)} \le r < c - d.$$

Then $N_2(X) \gg X/L^2$. Without the displayed non-A5 bound, the A5 route has not reached the direct endgame.

Proof. Use $S_X = R_5 + S_5$ and Corollary 4.2. □

6.4 Why the covariance hypothesis has not been proved by the notation

The genuine output primes satisfy

$$u = q_1 m + 2, \quad v = q_2 m + 2, \quad q_2 u - q_1 v = 2(q_2 - q_1).$$

This is a variable determinant shift. A theorem about primes in a single progression is not already a theorem about this binary correlation. The recent project lift $A = q_2 u$, $B = q_1 v$ gives

$$A - B = 2(q_2 - q_1), \qquad A - 2q_2 = B - 2q_1 = q_1 q_2 m.$$

The latter equalities are additional divisor locks. Ordinary averaged-shift theorems for fixed scalar sequences, such as [5], do not by their statements supply the locked estimate. This alternative remains a research proposal, not a premise silently discharged.

7 Optional universal Gram route

The normalized sequence is $w_X(n) = \Delta_X(n)/L$, the normalized prime comparison mass is $B_X^{\mathrm{norm}} = B_*(X)/L$, and the normalized exterior error is $E_H = S_X/L$. A Gram built from two copies of w_X differs by L^{-2} from the corresponding Gram for Δ_X.

Proposition 7.1 (Explicit conditional Gram export)**.** *Suppose that, for each requested $A > 0$, a source-exact reconstruction gives*

$$|E_H| \le K_A L^{d_A} \sup_{\omega\in\Omega_A(X)} |U_\omega| + K_A X L^{-A-2} + K_A X^{1-\xi_A} L^{d_A}, \qquad \xi_A > 0,$$

and that every actual generated row is

$$U_\omega = \sum_c \Gamma_\omega(c) \sum_m A_\omega(m) W_\omega(m,c) w_X(mc), \qquad M_\omega C_\omega \asymp X.$$

Assume, uniformly over the complete generated family,

$$\sum_c |\Gamma_\omega(c)|^2 \ll C_\omega L^{J_A}, \quad \sum_c \left| \sum_m A_\omega(m) W_\omega(m,c) w_X(mc) \right|^2 \ll_{A,N} X M_\omega L^{-N}$$

for every fixed $N > 0$. Then $E_H \ll_A X L^{-A}$ for every fixed $A > 0$, and $S_X = o(B_*(X))$.

Proof. Cauchy gives $|U_\omega|^2 \ll X^2 L^{J_A - N}$. Choose $N = 2A + 2d_A + J_A + 4$ and insert it into the reconstruction. The last power-saving error is $o(XL^{-A})$ for fixed parameters. Finally $S_X = LE_H$ and $B_*(X) \asymp X/L$. □

Both reconstruction and the family-wide Gram bound are hypotheses. A local A5 estimate, even if proved, does not establish them. This route is stronger than what the direct one-sided theorem needs. Ford–Maynard Proposition 7.22 assumes the relevant Type-II information; it cannot be read backwards as a proof of that information [1, pp. 54–60].

8 Verification ledger and remaining obligations

Item	Mathematical location	Status in this revision
Source and partition	Section 1; Lemma 2.1	Explicit definitions; exact proof supplied.
Comparison prime mass	Proposition 3.1	Proof supplied from classical maximal BV and truncation.
Complete physical Type I	Lemma 3.2; Proposition 3.3	Weighted, literal-comparison proof supplied; no exterior mask.
Interior positivity and leakage	Hypothesis 4.1	Not proved in full. The precise shrunken source and collar still require review.
Full exterior upper bound	Equation (4.2)	Open; not supplied by any finite check.
Rough transfer and moment	Hypotheses 5.1, 5.2	Optional analytic assumptions; not independently certified here.
Degree-twelve deduction	Proposition 5.3; Lemma 5.4	Proved conditional transfer and exact algebra.
Balanced reconstruction	Lemmas 6.1, 6.2	Fixed-collar proofs supplied; no claimed uniform shrinking-collar rate.
Dispersion background	Hypothesis 6.3	Supplied project proof retained for independent review.
Far prime-pair covariance	Hypothesis 6.5	Open. The target is $L^{-3-\eta}$ for the actual coefficient family.
Non-A5 aggregate	Corollary 6.7	Explicit additional obligation; not set to zero.
Universal physical export	Proposition 7.1	A separate conditional route, not required by the direct criterion.
Finite computation	Accompanying verifier and JSON	Reproducible regression checks only; no asymptotic or kernel certification.

Verification standard. The theorems proved here are ordinary mathematical deductions with stated inputs; they are not claims of external referee approval. The open hypotheses are fully written so that a specialist can check the implications without obtaining private project reports. Those private reports are still needed by anyone seeking to certify their proposed analytic proofs. No submitted paper, public repository, authorship record, or formal build is implied by this revision.

What is not a new result. The source weight is attributed to Ford and Maynard, the distribution and sieve inputs are classical, and the budget rearrangement is elementary. The sharper thresholds and the fixed-collar repair improve this manuscript's conditional formulation; they are not claimed to be historically novel or to prove twins.

A The finite A5 source data

The data below are the complete lower-index list from the supplied `BOX_CERTIFICATE.json`. Use weight 2 except for $(12, 15, 17, 25)$, which has weight 4. Indices divide by 100, with left-closed/right-open

intervals. This list is included to make the A5 statement externally identifiable; the main direct theorem does not need A5. The original certificate SHA-256 is

`42be326a87037aaa9f4a1e72c67fac39958a8165560443022442fd370f3f04d8`

(6, 8, 17, 29)	(6, 8, 18, 28)	(6, 8, 19, 27)	(6, 8, 20, 26)
(6, 8, 21, 25)	(6, 8, 22, 24)	(6, 9, 17, 29)	(6, 9, 18, 28)
(6, 9, 19, 27)	(6, 9, 20, 26)	(6, 9, 21, 25)	(6, 9, 22, 24)
(6, 10, 17, 29)	(6, 10, 18, 28)	(6, 10, 19, 27)	(6, 10, 20, 26)
(6, 10, 21, 25)	(6, 10, 22, 24)	(6, 11, 18, 28)	(6, 11, 19, 27)
(6, 11, 20, 26)	(6, 11, 21, 25)	(6, 11, 22, 24)	(6, 12, 19, 27)
(6, 12, 20, 26)	(6, 12, 21, 25)	(6, 12, 22, 24)	(6, 13, 20, 26)
(6, 13, 21, 25)	(6, 13, 22, 24)	(6, 14, 21, 25)	(6, 14, 22, 24)
(6, 15, 22, 24)	(7, 9, 17, 28)	(7, 9, 17, 29)	(7, 9, 18, 27)
(7, 9, 18, 28)	(7, 9, 19, 26)	(7, 9, 19, 27)	(7, 9, 20, 25)
(7, 9, 20, 26)	(7, 9, 21, 24)	(7, 9, 21, 25)	(7, 9, 22, 24)
(7, 10, 17, 28)	(7, 10, 18, 27)	(7, 10, 18, 28)	(7, 10, 19, 26)
(7, 10, 19, 27)	(7, 10, 20, 25)	(7, 10, 20, 26)	(7, 10, 21, 24)
(7, 10, 21, 25)	(7, 10, 22, 24)	(7, 11, 18, 27)	(7, 11, 19, 26)
(7, 11, 19, 27)	(7, 11, 20, 25)	(7, 11, 20, 26)	(7, 11, 21, 24)
(7, 11, 21, 25)	(7, 11, 22, 24)	(7, 12, 19, 26)	(7, 12, 20, 25)
(7, 12, 20, 26)	(7, 12, 21, 24)	(7, 12, 21, 25)	(7, 12, 22, 24)
(7, 13, 20, 25)	(7, 13, 21, 24)	(7, 13, 21, 25)	(7, 13, 22, 24)
(7, 14, 21, 24)	(7, 14, 22, 24)	(8, 10, 17, 27)	(8, 10, 18, 26)
(8, 10, 18, 27)	(8, 10, 19, 25)	(8, 10, 19, 26)	(8, 10, 19, 27)
(8, 10, 20, 24)	(8, 10, 20, 25)	(8, 10, 20, 26)	(8, 10, 21, 23)
(8, 10, 21, 24)	(8, 10, 21, 25)	(8, 10, 22, 24)	(8, 11, 18, 26)
(8, 11, 19, 25)	(8, 11, 19, 26)	(8, 11, 20, 24)	(8, 11, 20, 25)
(8, 11, 20, 26)	(8, 11, 21, 23)	(8, 11, 21, 24)	(8, 11, 21, 25)
(8, 11, 22, 24)	(8, 12, 19, 25)	(8, 12, 20, 24)	(8, 12, 20, 25)
(8, 12, 21, 23)	(8, 12, 21, 24)	(8, 12, 21, 25)	(8, 12, 22, 24)
(8, 13, 20, 24)	(8, 13, 21, 23)	(8, 13, 21, 24)	(8, 13, 22, 24)
(8, 14, 21, 23)	(9, 11, 18, 25)	(9, 11, 19, 24)	(9, 11, 19, 25)
(9, 11, 20, 23)	(9, 11, 20, 24)	(9, 11, 20, 25)	(9, 11, 21, 23)
(9, 11, 21, 24)	(9, 11, 21, 25)	(9, 11, 22, 24)	(9, 12, 19, 24)
(9, 12, 20, 23)	(9, 12, 20, 24)	(9, 12, 21, 23)	(9, 12, 21, 24)
(9, 12, 22, 24)	(9, 13, 20, 23)	(9, 13, 21, 23)	(10, 12, 19, 23)
(10, 12, 20, 22)	(10, 12, 20, 23)	(10, 12, 21, 23)	(10, 13, 20, 22)
(12, 15, 17, 25)			

The verifier also recomputes the original Ford coefficient, not just the stored weight label. On every box the first two primes are below n^{σ} and the last three are above n^{ν_0}, uniformly for $0 < \varepsilon \le \nu_0/100$. Expanding (1.5) reduces every support test to an affine inequality in $(x_1, x_2, x_3, x_4, \varepsilon)$ after substituting $x_5 = 1 - x_1 - x_2 - x_3 - x_4$. Exact extrema and endpoint-attainment flags on $[a_i, b_i)$ and $(0, \nu_0/100]$ determine each predicate on its full domain. No unresolved predicate is allowed. The resulting sum is 2 in 132 boxes and 4 in the exceptional box.

For exterior membership, every box has $\varepsilon < x_1 < \nu_0 - \varepsilon = \varepsilon + \sigma$. If its five-coordinate vector were a coagulation of a vector in $\mathscr{R}(P_\varepsilon)$, the coordinates producing x_1 would form a nonempty proper subsum in the forbidden interval, a contradiction. These are composite integers, so they belong to $\mathcal{E}_{\varepsilon,X}$. The script additionally checks balanced ranges, separated prime slots, and first-three selection uniqueness. None of this tests shifted primality.

B Repair log and scope of the finite checks

Preserved. The prime detector, comparison, shrunken g_ε, complete prime-power split, interior/exterior partition, exact endgame sign, proper-prime-power removal, degree-twelve expansion, and alternative Gram normalization are unchanged in mathematical content.

Improved conditional threshold. The old $c_F/4$ and $c_F/8$ targets were sufficient choices, not necessary gates. The general gap $c - t - d - s > 0$ exposes exactly how much is needed. After the explicit Type-I proof, $t = 0$ is available for fixed ε.

Newly self-contained supporting arguments. The comparison mass and literal weighted Type-I estimate no longer depend only on private filenames. Their proof is supplied with residues, main-density mismatch, divisor weights, bounded variation, and complete sieve-population costs.

Repaired latest-frontier rate. Fixed-domain boundary control does not automatically imply a uniform $O(\delta_L B_*(X))$ estimate for shrinking δ_L. Moreover such an error, even if true, can dominate a claimed $XL^{-1-\eta/2}$ main estimate. The repaired fixed-δ argument preserves the terminal conclusion $S_5 = o(B_*(X))$ but does not claim that sharper total rate.

Finite evidence. The included program verifies exact rational budget arithmetic, finite partition identities and inequalities, local comparison densities on complete CRT periods, Stirling expansions, projection facts, and source-data consistency. Its JSON records the counts and hashes from the actual run. It does not test an infinite asymptotic statement, independently certify the numerical c_F enclosure, or verify any universal R11 provider. The displayed proofs do not depend on the test count.

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References

[1] K. Ford and J. Maynard, *On the theory of prime-producing sieves*, arXiv:2407.14368v1 (2024). The cited numbering is that of the 107-page v1 PDF: Definitions 4.3, 4.5; Lemma 7.21; Proposition 7.22; equation (8.5); proof of Theorem 2.7(b). https://arxiv.org/abs/2407.14368v1.

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for the distinction between scalar average shifts and the project's unproved divisor-locked lift, not as a proof of Hypothesis 6.5. https://arxiv.org/abs/1707.01315.

[6] Supplied manuscript, *A Conditional Reduction of Twin-Prime Abundance to a Signed Ford-Weighted Exterior Estimate*, 27 October 2026, 20 pages; Twin_Primes_Conditional_Reduction_2026-10-27(2).pdf. Controlling source for the present revision; input hash is in the audit log.

[7] Supplied project records, 27 October 2026: R11_POST_FIRST_SUCCESS, R11_BALANCED_DISPERSION_REPAIR, and R11_TYPEII_TARGET_COMPRESSION, each with suffix _20261027, REPORT.md and companion files. Their claimed analytic results are used only at the explicit hypothesis or proved-adapter scope stated above. They are not public theorem citations.